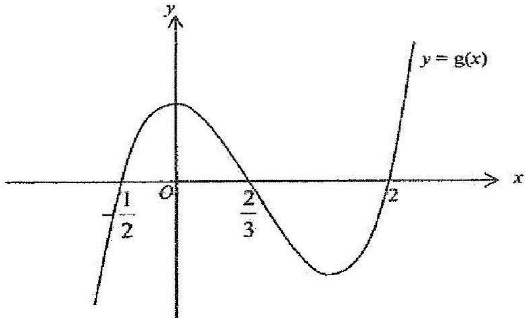
	Ans: (i) $a = 3, b = 2$ (ii) $x =- 2$
15.	(a) (i) Show that $(2x + 1)$ is a factor of $f(x)$ where $f(x) = 6x^3 - 13x^2 + 4$ *completely*. (ii) Given that $6x^3 - 13x^2 + 4 = (2x + 1)\left(ax^2 + bx + c\right)$, state the values of a, b and c. Factorize $6x^3 - 13x^2 + 4$ completely. (iii) The diagram shows the graph of $y = g(x)$ where $g(x)\neq f(x)$.  Write down a possible equation for the curve. (b) The polynomial $p(x)$ has a factor $(x - 3)$ and when divided by $(x + 1)$ leaves a remainder of 8. Given that $p(x) = (x + 1)(x - 3)q(x) + mx + c$, form two equations in m and c. Hence, find the remainder when $p(x)$ is divided by $\left(x^2 - 2x - 3\right)$. Ans: (a)(ii) $a = 3, b =- 8, c = 4, 6x^3 - 13x^2 + 4 = (2x + 1)(x - 2)(3x - 2)$ (iii) $m > 0, m\neq 1$ (b) $- 2x + 10$
16.	The expression $2x^3 + ax^2 + bx + 8$, where a and b are constants, has a factor of $x + 2$ and leaves a remainder of -24 when divided by $x - 2$. (i) Find the values of a and of b. (ii) Using the values of a and of b found in part (i), solve the equation $2x^3 + ax^2 + bx + 8 = 0$. Ans: (i) $a =- 5, b =- 14$ (ii) -2 or $\frac{1}{2}$ or 4
17.	The term containing the highest power of x in the polynomial $f(x)$ is $- 2x^4$. Two of the roots of the equation $P(x) = 0$ are -1 and 2 . The third factor of $f(x)$ is $\left(x^2 + mx - m\right)$.

	(a) Write down, without expanding, an expression for the polynomial $f(x)$. (b) In the case where all the roots of the equation $f(x) = 0$ are real, find the range of values of m. (c) In the case where $m =- 4$, find the number of distinct real roots of the equation $f(x) = 0$. (d) In the case where $m =- 1$, explain using your answer in (b) or otherwise, why $\left(x^2 + mx - m\right)$ is always positive. Hence solve the inequality $f(x) > 0$. Ans: (a) $- 2(x + 1)(x - 2)\left(x^2 + mx - m\right)$ (b) $m\leq - 4$ or $m\geq 0$ (c) 2(d) $- 1 < x < 2$
18.	$f(x) = 6x^3 + ax^2 + bx - 6$ has a factor $x + 2$ but leaves a remainder of -12 when divided by $x - 1$. (i) Find the value of a and of b. (ii) Factorise $f(x)$ completely and hence solve the equation $48x^3 + 4ax^2 = 6 - 2bx$. Ans: (i) $a = 5, b =- 17$ (ii) $f(x) = (x + 2)(3x - 1)(2x - 3), x =- 1, -\frac{1}{6}, \frac{3}{4}$
19.	The expression $px^3 - 3qx^2 + qx + 2p$ is exactly divisible by ($2x + 1$) and has a remainder of -10 when divided by $(x + 1)$. (a) Show that the values of p and q are 2 and 3 respectively. (b) Hence, or otherwise, (i) factorise the above expression completely, and (ii) determine the remainder when the expression above is divided by $(x - 3)$. Ans: (b)(i) $(2x + 1)(x - 4)(x - 1)$ (ii) -14
20.	(i) Given that $f(x) = 2x^3 + ax^2 + bx - 30$ has a factor $(x + 3)$ and leaves a reminder of -28 when divided by $(x - 1)$. Find the values of a and of b and solve $f(x) = 0$. (ii) Hence solve $2(y + 1)^3 + a(y + 1)^2 + by + b - 30 = 0$. Ans: $a = 7, b =- 7; x =- 3, 2$ or -2.5 (ii) $y =- 4, 1$ or -3.5
21.	Given the cubic expression $f(x) = x^3 + px^2 + qx + 4$ has a factor $(x + 2)$ and leaves a remainder of 6 when divided by $(x + 1)$, (i) find the value of p and of q,

	(ii) factorize $f(x)$ completely. Ans: (i) $p =- 1, q =- 4$ (ii) $f(x) = (x + 2)(x - 2)(x - 1)$
22.	The cubic polynomial $f(x)$ is such that the coefficient of x^3 is 2 and the roots of the equation $f(x) = 0$ are $2, -\frac{1}{2}$ and k. Given that $f(x)$ has a remainder of -6 when divided by $x - 1$. Find the value of k. Ans: $k =- 1$
23.	The expression $f(x) = 2x^3 + (2b - a)x^2 - bx + c$ is exactly divisible by $2x^2 - x$. When $f(x)$ is divided by $x + 1$, the remainder is 6 . Find the value of a, of b and of c. Hence factorise $f(x)$ completely. Ans: $f(x) = x(x + 3)(2x - 1)$
24.	It is given that $x + 4$ is a factor of $g(x) + 3$, where $g(x)$ is a polynomial. Find the remainder when $f(x) = \left(x^3 - 3x + 2\right)g(x)$ is divided by $x + 4$. Ans: 150
25.	Given that $f(x) = (3x - 5)^2(x + 3) - (x - 9)^2$, (i) find the remainder when $f(x)$ is divided by $x + 1$. (ii) show that $x + 2$ is a factor of $f(x)$ and hence solve the equation $f(x) = 0$, giving your answers correct to 2 decimal places where necessary. Ans: (i) 28 (ii) $x =- 2$ or -0.13 or 2.57
26.	The function $f(x) = 1 + 2x + Ax^2 - x^3$, where A is a constant, leaves a remainder of $1\frac{3}{8}$ when divided by $(2x - 1)$. (i) Find the value of A. (ii) Hence solve the equation $f(x) = 0$, giving your answers in the exact form. Ans: (i) -2 (ii) $x = 1, \frac{-3\pm\sqrt{5}}{2}$
27.	The function $f(x)$ is such that $f(x) = 2x^3 + 3x^2 - x - 4$, (i) find a factor of $f(x)$. **(ii)** Hence, determine the number of solutions in the equation $f(x) = 0$. Ans: (i) $(x - 1)$ (ii) one solution